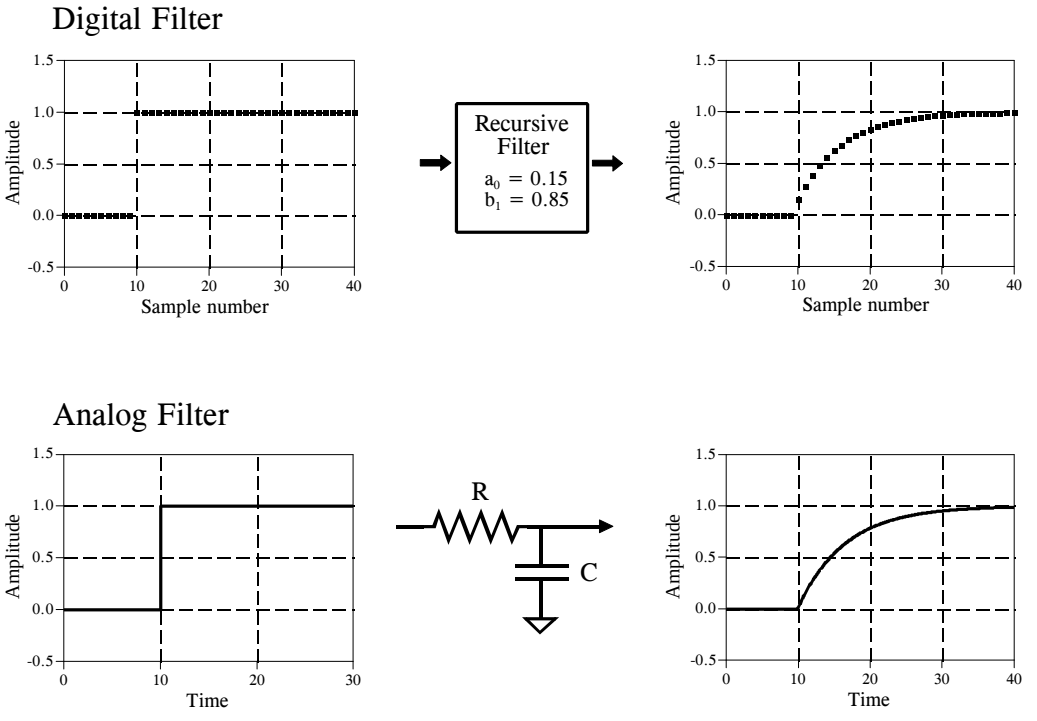

FIGURE 19-2
Single pole low-pass filter. Digital recursive filters can mimic analog filters composed of resistors and capacitors. As shown in this example, a single pole low-pass recursive filter smoothes the edge of a step input, just as an electronic RC filter.

Single Pole Recursive Filters

Figure 19-2 shows an example of what is called a **single pole** low-pass filter. This recursive filter uses just two coefficients, $a_0 = 0.15$ and $b_1 = 0.85$. For this example, the input signal is a step function. As you should expect for a low-pass filter, the output is a smooth rise to the steady state level. This figure also shows something that ties into your knowledge of electronics. This low-pass recursive filter is completely analogous to an electronic low-pass filter composed of a single resistor and capacitor.

The beauty of the recursive method is in its ability to create a wide variety of responses by changing only a few parameters. For example, Fig. 19-3 shows a filter with three coefficients: $a_0 = 0.93$, $a_1 = -0.93$ and $b_1 = 0.86$. As shown by the similar step responses, this digital filter mimics an electronic RC high-pass filter.

These single pole recursive filters are definitely something you want to keep in your DSP toolbox. You can use them to process digital signals just as you would use RC networks to process analog electronic signals. This includes everything you would expect: DC removal, high-frequency noise suppression, wave shaping, smoothing, etc. They are easy to program, fast